

- 18** A isolated metal sphere of radius 0.1 m is positively charged. A small charge was brought from a distant point to a point 0.5 m from the centre of the metal sphere. The work done against the electric field is W . At its final position, the electric force on the charge is F .

If the charge has been brought to a point 1.0 m from the centre of the metal sphere, what would have been the values for the work done against the electric field and the electric force on the charge at its final position?

	work done against electric field	force on charge at its final position
A	$\frac{W}{4}$	$\frac{F}{4}$
B	$\frac{W}{4}$	$\frac{F}{2}$
C	$\frac{W}{2}$	$\frac{F}{4}$
D	$\frac{W}{2}$	$\frac{F}{2}$

- 19** The electrical potential difference between two points in a wire carrying a current is